

Early Clinical and EEG Association of Genotype and Outcome in Genetic Epilepsies

A Cohort Study and Hierarchical Clustering Analysis

Angela De Dominicis,^{1,*} Mattia Mercier,^{1,2,*} Giusy Carfi Pavia,¹ Ludovica Maria Piscitello,¹ Simona Cappelletti,¹ Cristina Fillosomi,¹ Cinzia Correale,³ Costanza Calabrese,¹ Luca De Palma,¹ Antonio Novelli,⁴ Fabrizia Stregapede,⁴ Marina Trivisano,¹ and Nicola Specchio^{1,5}

Correspondence

Dr. Specchio
nicola.specchio@opbg.net

Neurology® 2026;106:e214761. doi:10.1212/WNL.00000000000214761

Abstract

Background and Objectives

Genetic epilepsies include a broad spectrum of disorders caused by pathogenic variants in more than 1,000 genes. Their clinical expression is highly variable, making early phenotype-genotype interpretation challenging. Early seizure semiology and EEG features may offer clinically useful information for diagnostic orientation and management. The aim of this study was to characterize early clinical and EEG features in patients with genetic epilepsies, examine their associations with outcomes, and explore genotype-phenotype groupings through hierarchical clustering analysis (HCA).

Methods

We conducted a retrospective study at Bambino Gesù Children's Hospital. Eligible participants carried pathogenic or likely pathogenic variants in epilepsy-related genes, identified through medical records and laboratory diagnostic logs. Clinical variables at seizure onset and EEG recordings performed within the first month of the initial seizure were extracted. Follow-up outcomes included seizure frequency, drug resistance, movement disorders, behavioral/autism spectrum disorder comorbidities, and developmental delay/intellectual disability (DD/ID). Associations between early features and outcomes were assessed using χ^2 or Fisher tests. HCA was used to identify clusters linking early phenotype and gene-level etiology.

Results

We included 277 patients (52.3% female; median age at last follow-up 8.1 years, range 0–40). Drug resistance occurred in 58.8% and severe DD/ID in 35.4% of patients. EEG data at onset were available for 107 individuals. Neonatal onset was associated with a higher rate of drug resistance (71.4%; odds ratio [OR] 2.0, 95% CI 1.05–3.77), movement disorders (60.7%; OR 3.7, 95% CI 2.02–6.82), and severe DD/ID (71.4%; OR 7.0, 95% CI 3.66–13.49). Slow EEG background activity and multifocal epileptiform discharges were associated with both drug resistance and severe DD/ID. HCA identified genotype-phenotype groupings, including clusters involving *SCN1A*, *PRRT2*, *STXBP1*, *KCNQ2*, *SCN2A*, *CHD2*, *SYNGAP1*, and *MECP2*, each linked to specific clinical and EEG features.

Discussion

Early clinical and EEG features showed meaningful associations with outcomes and mapped onto specific genetic etiologies. HCA revealed coherent genotype-phenotype clusters that may support early diagnostic reasoning. Limitations include the retrospective design and small numbers per gene, warranting larger multicenter studies for validation.

RELATED ARTICLE

Editorial

Early Electroclinical Phenotypes of Genetic Epilepsies Identified Through Unsupervised Machine Learning: AI in the House of Gastaut
Page e214884

*These authors contributed equally to this work.

¹Neurology, Epilepsy and Movement Disorders Unit, Bambino Gesù Children's Hospital, IRCCS, Full Member of European Reference Network EpiCARE, Rome, Italy; ²Department of Physiology, Behavioral Neuroscience PhD Program, Sapienza University, Rome, Italy; ³Centre for Behavioral Sciences and Mental Health, Istituto Superiore di Sanità, Rome, Italy; ⁴Laboratory of Medical Genetics, Translational Cytogenomics Research Unit, Bambino Gesù Children's Hospital, IRCCS, Rome, Italy; and ⁵University Hospitals KU Leuven, Belgium.

Glossary

ASD = autism spectrum disorder; **ASO** = antisense oligonucleotide; **BA** = background activity; **DD** = developmental delay; **DEE** = developmental and epileptic encephalopathy; **DS** = Dravet syndrome; **EMaTS** = epilepsy with myoclonic-atonic seizure; **GEFS+** = genetic epilepsy with febrile seizures plus; **HCA** = hierarchical clustering analysis; **ID** = intellectual disability; **LOF** = loss of function; **OR** = odds ratio; **SeLIE** = self-limited infantile epilepsy; **SeLNE** = self-limited neonatal epilepsy.

Introduction

Genetic epilepsies affect over 0.4% of the general population and are responsible for more than 30% of all epilepsies,¹ while in the past, the etiology was unknown in approximately 75% of patients. Advances in next-generation sequencing led to the identification of an increasing number of genes involved in neuronal excitability, synaptic signaling, and early brain development. To date, more than 1,000 epilepsy genes have been identified.^{2–4} Sequence changes have been identified in 30%–40%, while copy number variants are responsible for 5%–12% of genetic epilepsies.⁵

Every genetic etiology can be associated with different epileptic phenotypes and outcomes, ranging from self-limited epilepsies to developmental and epileptic encephalopathies (DEEs).

Individuals carrying pathogenic variants in the same gene may exhibit marked differences in the age at onset, seizure types, developmental trajectories, and comorbidity profiles. Conversely, similar clinical presentations can arise from distinct genetic etiologies. Indeed, the same gene can be associated with a wide range of phenotypes, from self-limited epilepsies to DEEs as seen in ion channel genes⁶ (*KCNQ2*, *SCN1A*, *SCN2A*, and *SCN8A*). For example, loss-of-function (LOF) *KCNQ2* genetic variants can cause self-limited neonatal epilepsy (SeLNE), in which seizures develop on days 2–3 of life and outcome are excellent, with only a small proportion of patients developing late epilepsy.⁷ In vitro studies demonstrated haploinsufficiency as a likely disease mechanism.⁸ On the contrary, *KCNQ2* genetic variants are also associated with a severe neonatal-onset DEE characterized by tonic seizures and profound developmental impairment.⁹ These patients often have de novo genetic variants, which may be dominant negative, with a more profound overall negative effect on channel function than those responsible for SeLNE.⁹ Another example is *SCN1A*, causing both Dravet syndrome (DS), overall in patients harboring LOF genetic variants, and genetic epilepsy with febrile seizures plus (GEFS+), a mild self-limited epilepsy most frequently in the presence of missense variants.¹⁰ This phenotypic variability complicates prognostication, treatment decisions, and family counseling and has underscored the need for refined stratification approaches.

Phenotypic heterogeneity in genetic epilepsies arises from multiple factors, including the type of genetic variant, the developmental timing of gene expression, mosaicism, epigenetic

mechanisms, and modifier genes.¹¹ Moreover, common genetic variants and environmental influences may modulate the effects of rare pathogenic variants, further amplifying variability among individuals with the same genotype.¹² Although the identification of modifier factors could, in the future, improve prognostic prediction, these investigations are not yet available in clinical practice. As a result, clinicians are often faced with both genetic and phenotypic heterogeneity at the onset of epilepsy. Recognizing early clinical and EEG markers that may suggest a specific causative gene is, therefore, crucial to optimize molecular testing strategies and guide both diagnostic and therapeutic decisions.

The aim of this study was to explore the correlation between clinical and EEG data at epilepsy onset and outcome, in patients with genetic epilepsies. We conducted a hierarchical clustering analysis (HCA) to identify homogeneous subgroups of causative genes in which specific clinical and EEG patterns were identified.

Methods

Sample Collection

The analysis follows an association-based and causal-interpretation framework, consistent with STrengthening the REporting of Genetic Association Studies reporting guidelines.¹³

Between 2014 and 2022, we performed genetic examinations in approximately 1,000 individuals with epilepsy at the Neurology, Epilepsy and Movement Disorders Unit of Bambino Gesù Children's Hospital (Rome, Italy). Our center is a national referral hub for rare and complex epilepsies, receiving patients from across Italy and from international collaborators. We do not serve a geographically defined population; instead, patients are referred for specialized diagnostic evaluation, second opinions, presurgical workup, or management of DEE. Not all patients with epilepsy seen at our center undergo genetic testing. Genetic testing is recommended based on predefined clinical criteria, including the following: (1) epilepsy onset in the neonatal period or first year of life; (2) drug-resistant epilepsy; (3) diagnosis of DEE; (4) epilepsy associated with developmental delay (DD), intellectual disability (ID), autism spectrum disorder (ASD), or movement disorders; and (5) familial epilepsy suggestive of a monogenic cause. Genetic analyses include targeted epilepsy gene panels, whole-exome sequencing, and copy number

variant assessment. To assemble the cohort for this study, we screened medical records and laboratory diagnostic logs to identify all consecutive individuals carrying genetic variants classified as likely pathogenic or pathogenic according to American College of Medical Genetics and Genomics criteria. Patients in our cohort had a wide spectrum of epilepsies, from self-limited to DEEs.

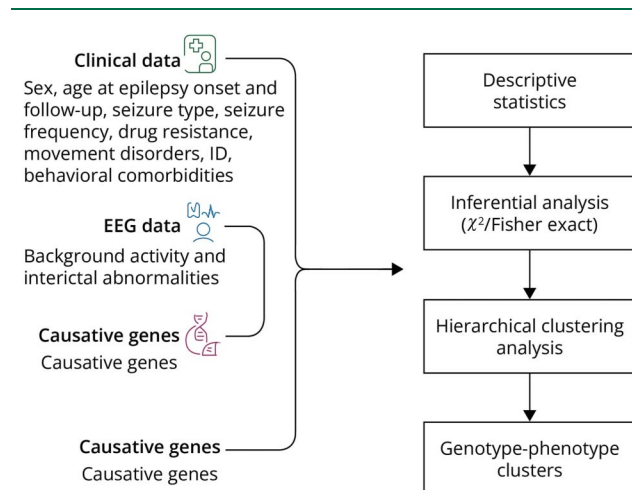
Patients were routinely followed at our center with clinical, EEG, and neurodevelopmental evaluations every 6 months. We retrospectively collected the following data for each patient: (1) Clinical data: sex, age at epilepsy onset (categorized into 3 mutually exclusive groups: “neonatal,” defined as onset within the first 28 days of life; “within the 1st year of life,” defined as onset between day 29 and 12 months of age; and “after the 1st year of life”), age at last follow-up (less or more than 5 years), type of seizures at onset (febrile or afebrile, focal, generalized, combined [both focal and generalized seizures], or infantile epileptic spasms). At the last follow-up, we collected the following data: seizure frequency, which was classified as daily-weekly, monthly, sporadic (at least 1 seizure per year), and seizure-free (at least for 2 years); presence of drug resistance; behavioral comorbidities; and/or ASDs, movement disorders, and DD/ID (none, mild, moderate, or severe). Seizure frequency was categorized as daily, weekly, monthly, sporadic, or seizure-free (≥ 2 years). In this study, “sporadic” was defined as a seizure frequency ranging from 1 to 11 seizures per year, that is, at least 1 seizure annually but not reaching the threshold for monthly occurrence. Patients with sporadic seizures could still be classified as drug resistant if they fulfilled the ILAE definition, namely persistence of seizures despite adequate trials of at least 2 appropriately chosen antiseizure medications. (2) EEG at epilepsy onset (available for 107 of 277 patients): we evaluated background activity (BA) and stratified as normal, slow, or other (hypsarhythmia, burst suppression, or paroxysmal alternating pattern), and the presence of interictal epileptiform abnormalities (no abnormalities, focal/multifocal, or generalized). EEG recordings at onset were obtained through standard clinical monitoring lasting at least 2 hours, using a standard montage. (3) Causative genes.

The 3 domains of data used for the analysis are illustrated in Figure 1.

In this study, “at onset” was defined as clinical and EEG information obtained within the first month after the first seizure. Clinical features were derived from the characterization of the first seizure. EEG features were extracted from the first standard 2-hour EEG monitoring conducted within this time frame. In patients in whom multiple EEGs were available during the first month, the earliest was used for the analysis.

All continuous or discrete variables were transformed into categorical variables (dichotomous). This transformation was undertaken for simplified descriptive and statistical/clustering analysis, making it more manageable and interpretable.

Figure 1 Representation of the Data Domains Collected for Each Patient



Schematic representation of the 3 main data domains collected for each patient: (1) clinical data (sex, age at epilepsy onset and follow-up, seizure type, seizure frequency, drug resistance, movement disorders, intellectual disability (ID), and behavioral comorbidities), (2) EEG data at seizure onset (background activity and interictal abnormalities), and (3) causative genes. These inputs were first explored using descriptive statistics, followed by inferential tests (χ^2 and Fisher exact) to assess associations between features and clinical outcomes. Hierarchical clustering analysis was performed to integrate clinical, EEG, and genetic data, enabling the identification of genotype-phenotype clusters. The diagram highlights that all 3 domains jointly contributed to the analytical pipeline, rather than reflecting a chronological order of acquisition. Causative genes were as follows: *SCN1A* (57), *KCNQ2* (23), *PCDH19* (20), *SCN2A* (17), *SCN8A* (15), *CDKL5* (13), *STXBP1* (12), *PRRT2* (11), *CHD2* (10), *GRIN2A* (6), *KCNB1* (6), *GABRB3* (6), *MECP2* (4), *WVVOX* (4), *CSNK2B* (5), *SYNGAP1* (5), *ALG13* (2), *EEF1A2* (2), *GABRA5* (2), *GRIN2D* (2), *SCN1B* (2), *SLC13A5* (2), *SLC6A1* (2), *SZT2* (2), *DEPDC5* (4), *GABRA1* (4), *KCNA2* (3), *KCNT1* (3), *PIGA* (3), *DNM1L* (3), *BRAT1* (1), *CAMK2A* (1), *CAMK2B* (1), *CDK19* (1), *CNKSR2* (1), *FGF12* (1), *GABRB2* (1), *GRIA3* (1), *GRIN1* (1), *HUWE1* (1), *KCNA1* (1), *KCNK2* (1), *KCNH2* (1), *KCNK4* (1), *MBOAT7* (1), *MED13* (1), *NEXMIF* (1), *PIGN* (1), *SLC2A1* (1), *SPTAN1* (1), *STX1B* (1), *TBC1D24* (1), and *TRIO* (1).

Statistical Analysis

For all statistical analyses, we used the Statistical Package for the Social Sciences software. Instead, for clustering computation, we used MATLAB (2022b, Mathworks, Natick, MA).

To gain comprehensive insights into the data set, we conducted a descriptive statistical analysis, including frequency distributions and the summary of continuous variables through measures such as mean, median, standard deviation, and range min/max.

We performed the χ^2 test and Fisher exact test. We aimed to assess the significance of associations between clinical and/or electrical data at onset and the variables at follow-up or the specific genetic etiologies.

Hierarchical Clustering Analysis

We conducted 3 HCAs to explore the underlying structure of the data set and identify natural groupings within the data. This method was chosen because of its ability to generate a hierarchical representation of relationships between different variable combinations, facilitating the identification of clinically relevant clusters. The clustering models applied were as follows:

- Clinical features at onset vs causative genes
- EEG features at onset vs causative genes
- Clinical and EEG features at onset vs causative genes

Three dissimilarity correlation matrices were computed, which were used to perform automatic hierarchical cluster analysis. The Kendall coefficient was used to assess the correlation in terms of both strength and direction between variables. We excluded patients whose causative gene frequency was below 4. This exclusion was implemented to ensure the reliability and robustness of our statistical findings, as small sample sizes can lead to unstable and potentially biased results.

Standard Protocol Approvals, Registrations, and Patient Consents

Patients (or legal guardians) provided consent for genetic testing and use of anonymized data. This study was performed according to ethical principles for medical research involving human participants stated in the Declaration of Helsinki and was approved by the local ethics committee (protocol no. 3496_OPBG_2024).

Data Availability

Anonymized data not published within this article will be made available on request from any qualified investigator.

Results

Among the whole sample of patients who underwent genetic testing, 277 patients (132 [47.7%] male) carried a pathogenic or likely pathogenic variant and were included in the study. The median age at enrollment was 9 ± 7.74 years (range 0–40). Demographic, clinical, and EEG data are summarized in Table 1.

One hundred forty-nine patients (53.8%) had seizure onset within the first year of life. Focal seizures were the most common (53.4%), and they were typically afebrile (75.1%). At follow-up, 163 of 277 (58.8%) had drug-resistant epilepsy and 93 of 277 (33.6%) experienced sporadic seizures of whom 51 (55%) had drug resistant epilepsy, 99 (35.7%) had movement disorders, and 115 (41.5%) had behavioral/ASD comorbidities. Furthermore, 98 patients (35.4%) exhibited severe DD/ID. Follow-up duration was longer than 5 years in 174 patients (62.8%). EEG data were available for 107 of 277 (38.6%): most patients exhibited normal BA (60.7%), and most patients did not have interictal epileptiform abnormalities (48.6%).

Results of statistical analysis are summarized in Table 2. Age at onset showed associations with follow-up variables. Neonatal onset was followed by daily seizures in 46.7%, whereas onset after the first year was more frequently associated with sporadic seizures (48.0%; *p* = 0.046). Drug resistance occurred in 73.3% of patients with neonatal onset compared with 58.2%

Table 1 Demographic, Clinical, and EEG Data

	Number (%)
Total number of patients	277
Sex	
Male	132 (47.7)
Female	145 (52.3)
Age at onset	
Neonatal	56 (20.2)
Within first year of life	149 (53.8)
After first year of life	72 (26)
Type of seizure	
Focal	148 (53.4)
Generalized	96 (34.7)
Combined	16 (5.8)
Epileptic spasms	17 (6.1)
Seizure occurrence	
Febrile	69 (24.9)
Afebrile	208 (75.1)
Seizure frequency at follow-up	
Seizure-free	75 (27.1)
Daily/weekly	83 (30)
Monthly	26 (9.4)
Sporadic	93 (33.6)
Drug resistance	
No	114 (41.2)
Yes	163 (58.8)
Movement disorders	
No	178 (64.3)
Yes	99 (35.7)
Behavioral/ASD symptoms	
No	162 (58.8)
Yes	115 (41.5)
Intellectual disability	
No	86 (31)
Mild	40 (14.4)
Moderate	53 (19.1)
Severe	98 (35.4)
Age at follow-up	
≤5 y	103 (37.2)
≥5 y	174 (62.8)

Continued

Table 1 Demographic, Clinical, and EEG Data (continued)

	Number (%)
Patients with EEG data available	107
Background EEG activity	
Normal	65 (60.7)
Slow	28 (26.2)
Other	14 (13.1)
Epileptiform abnormalities	
Focal/multifocal	43 (40.2)
Generalized	12 (11.2)
None	52 (48.6)

Abbreviation: ASD = autism spectrum disorder.

for onset within the first year and 32.0% after 1 year ($p = 0.012$). Movement disorders were more common in neonatal-onset patients (53.3%) than in those with onset within the first year (29.9%) or after 1 year (24.0%; $p < 0.001$). Severe DD/ID occurred in 60% of patients with neonatal onset vs 32.8% and 28.0% in the other 2 groups ($p < 0.001$).

Seizure type was also associated with outcomes. Movement disorders were more frequent in patients with combined seizures (60.0%) or epileptic spasms (77.8%) than in those with focal (20.0%) or generalized (34.2%) seizures ($p = 0.007$). Severe DD/ID was most common in patients with epileptic spasms (77.8%) and combined seizures (40.0%), compared with focal (29.1%) and generalized (34.2%) seizures ($p < 0.001$).

Febrile vs afebrile seizures showed distinct patterns. Sporadic seizures occurred in 45.8% of patients with febrile seizures vs 31.3% with afebrile seizures, whereas weekly seizures occurred in 4.2% and 7.2%, respectively ($p = 0.005$). Drug resistance was less frequent in patients with febrile than in those with afebrile seizures (41.7% vs 57.8%; $p = 0.015$). Movement disorders were present in 8.3% of patients with febrile seizures and 38.6% with afebrile seizures ($p < 0.001$). Severe DD/ID occurred in 8.3% of patients with febrile vs 43.4% of patients with afebrile seizures ($p < 0.001$).

Drug resistance occurred in 67.9% of patients with slow BA vs 43.1% with normal BA ($p = 0.013$), severe DD/ID in 46.4% vs 23.1% ($p = 0.019$), and the presence of movement disorders in 39.3% vs 21.5% ($p = 0.005$).

Drug resistance occurred in 72.1% of patients with focal/multifocal epileptiform abnormalities vs 38.5% of those without abnormalities ($p = 0.004$), whereas severe DD/ID occurred in 55.8% vs 23.1% ($p < 0.001$). Focal/multifocal epileptiform abnormalities were associated with 48.8% of

patients with movement disorders vs 21.2% without ($p = 0.008$).

Hierarchical Clustering Analysis

Clinical Features at Onset vs Causative Gene

We performed HCA to explore the correlation coefficient between clinical features at onset and causative genes in 226 of 277 patients. Matrix correlation showed that a specific cluster of genes composed of *SYNGAP1* and *CHD2* was positively correlated with generalized seizures ($r = 0.31$, $r = 0.33$) and epilepsy onset after 1 year ($r = 0.22$, $r = 0.26$). *SCN1A* was positively correlated with febrile seizure ($r = 0.58$) and onset within 1 year ($r = 0.42$). In addition, we found that a cluster of genes composed of *SCN2A* and *KCNQ2* were positively associated with neonatal epilepsy onset ($r = 0.31$, $r = 0.48$). The statistical correlation between each combination of clinical features at onset and causative gene is presented in Figure 2.

EEG Features at Onset vs Causative Gene

We performed HCA to explore the correlation coefficient between EEG data at onset and causative genes in 61 of 107 patients with EEG data available at onset (Figure 3). Results revealed that *SCN1A* and *PRRT2* exhibited positive correlations with a cluster of EEG data characterized by normal BA ($r = 0.36$, $r = 0.17$) and the absence of epileptiform abnormalities ($r = 0.29$, $r = 0.36$). In addition, we identified significant correlations between *STXBP1* and the presence of burst suppression/hypsarrhythmia/paroxysmal alternating background EEG activity ($r = 0.47$), as well as focal/multifocal epileptiform abnormalities ($r = 0.37$).

Clinical and EEG Features at Onset vs Causative Gene

In the context of an integrated analysis of clinical and EEG data, our findings confirmed previously reported correlations and uncovered novel associations. The occurrence of *SCN1A* genetic variants was strongly associated with a specific cluster of clinical and EEG data, including febrile seizures ($r = 0.57$), epilepsy onset within the first year of life ($r = 0.39$), normal BA ($r = 0.36$), and the absence of epileptiform abnormalities ($r = 0.29$).

PRRT2 remained positively correlated with normal BA ($r = 0.17$) and the absence of epileptiform abnormalities ($r = 0.36$), as previously mentioned. Moreover, *PRRT2* had a positive correlation with epilepsy onset within the first year of life ($r = 0.29$) and an independent correlation with afebrile seizures ($r = 0.27$) and focal seizures ($r = 0.22$).

STXBP1 was positively correlated with a cluster including infantile epileptic spasms ($r = 0.51$) and the presence of burst suppression/hypsarrhythmia/paroxysmal alternating BA ($r = 0.47$), along with an independent correlation with focal/multifocal epileptiform abnormalities ($r = 0.37$).

KCNQ2 and *SCN2A* were both associated with clusters characterized by neonatal epilepsy onset ($r = 0.42$, $r = 0.37$,

Table 2 Statistical Analysis Between Clinical and EEG Data at Onset and at Follow-Up

Clinical and EEG data at onset	Clinical and EEG data at follow-up									
	Seizure frequency, n (%)	p Value	DR, n (%)	p Value	MD, n (%)	p Value	Behavior/ASD, n (%)	p Value	DD/ID, n (%)	p Value
Age at onset										
Neonatal (n = 15)	Sporadic: 1 (6.7) Daily: 7 (46.7) Free: 7 (46.7)	0.046*	No: 4 (26.7) Yes: 11 (73.3)	0.012*	No: 7 (46.7) Yes: 8 (53.3)	0.000*	No: 13 (86.7) Yes: 2 (13.3)	0.000*	Moderate: 2 (13.3) Severe: 9 (60.0) None: 4 (26.7)	0.000*
Within 1 y (n = 67)	Sporadic: 24 (35.8) Monthly: 10 (14.9) Weekly: 4 (6.0) Daily: 14 (20.9) Free: 15 (22.4)		No: 28 (41.8) Yes: 39 (58.2)		No: 47 (70.1) Yes: 20 (29.9)		No: 46 (68.7) Yes: 21 (31.3)		Mild: 14 (20.9) Moderate: 9 (13.4) Severe: 22 (32.8) None: 22 (32.8)	
After 1 y (n = 25)	Sporadic: 12 (48.0) Weekly: 3 (12.0) Daily: 4 (16.0) Free: 6 (24.0)		No: 17 (68.0) Yes: 8 (32.0)		No: 19 (76.0) Yes: 6 (24.0)		No: 14 (56.0) Yes: 11 (44.0)		Mild: 4 (16.0) Moderate: 4 (16.0) Severe: 7 (28.0) None: 10 (40.0)	
Type of seizures										
Combined (n = 5)	Sporadic: 1 (20.0) Monthly: 1 (20.0) Daily: 2 (40.0) Free: 1 (20.0)	0.156	Yes: 5 (100.0)	0.192	No: 2 (40.0) Yes: 3 (60.0)	0.007*	No: 2 (40.0) Yes: 3 (60.0)	0.182	Mild: 2 (40.0) Moderate: 1 (20.0) Severe: 2 (40.0)	0.000*
Focal (n = 55)	Sporadic: 17 (30.9) Monthly: 6 (10.9) Weekly: 1 (1.8) Daily: 13 (23.6) Free: 18 (32.7)		No: 29 (52.7) Yes: 26 (47.3)		No: 44 (80.0) Yes: 11 (20.0)		No: 43 (78.2) Yes: 12 (21.8)		Mild: 7 (12.7) Moderate: 8 (14.5) Severe: 16 (29.1) None: 24 (43.6)	
Generalized (n = 38)	Sporadic: 14 (36.8) Monthly: 2 (5.3) Weekly: 6 (15.8) Daily: 9 (23.7) Free: 7 (18.4)		No: 16 (42.1) Yes: 22 (57.9)		No: 25 (65.8) Yes: 13 (34.2)		No: 24 (63.2) Yes: 14 (36.8)		Mild: 8 (21.1) Moderate: 5 (13.2) Severe: 13 (34.2) None: 12 (31.6)	
Spasms (n = 9)	Sporadic: 5 (55.6) Monthly: 1 (11.1) Daily: 1 (11.1) Free: 2 (22.2)		No: 4 (44.4) Yes: 5 (55.6)		No: 2 (22.2) Yes: 7 (77.8)		No: 4 (44.4) Yes: 5 (55.6)		Mild: 1 (11.1) Moderate: 1 (11.1) Severe: 7 (77.8)	
Febrile/afebrile seizure										
Febrile (n = 24)	Sporadic: 11 (45.8) Monthly: 7 (29.2) Weekly: 1 (4.2) Free: 5 (20.8)	0.005*	No: 14 (58.3) Yes: 10 (41.7)	0.015*	No: 22 (91.7) Yes: 2 (8.3)	0.000*	No: 16 (66.7) Yes: 8 (33.3)	0.220	Mild: 7 (29.2) Moderate: 4 (16.7) Severe: 2 (8.3) None: 11 (45.8)	0.000*
Afebrile (n = 83)	Sporadic: 26 (31.3) Monthly: 3 (3.6) Weekly: 6 (7.2) Daily: 25 (30.1) Free: 23 (27.7)		No: 35 (42.2) Yes: 48 (57.8)		No: 51 (61.4) Yes: 32 (38.6)		No: 57 (68.7) Yes: 26 (31.3)		Mild: 11 (13.3) Moderate: 11 (13.3) Severe: 36 (43.4) None: 25 (30.1)	
Background EEG activity										
Normal (n = 65)	Sporadic: 20 (30.8) Monthly: 9 (13.8) Weekly: 6 (9.2) Daily: 10 (15.4) Free: 20 (30.8)	0.233	No: 37 (56.9) Yes: 28 (43.1)	0.013*	No: 51 (78.5) Yes: 14 (21.5)	0.005*	No: 44 (67.7) Yes: 21 (32.3)	0.638	Mild: 14 (21.5) Moderate: 10 (15.4) Severe: 15 (23.1) None: 26 (40.0)	0.019*

Continued

Table 2 Statistical Analysis Between Clinical and EEG Data at Onset and at Follow-Up (continued)

Clinical and EEG data at onset	Clinical and EEG data at follow-up									
	Seizure frequency, n (%)	p Value	DR, n (%)	p Value	MD, n (%)	p Value	Behavior/ASD, n (%)	p Value	DD/ID, n (%)	p Value
Other (n = 14)	Sporadic: 7 (50.0) Daily: 5 (35.7) Free: 2 (14.3)		No: 3 (21.4) Yes: 11 (78.6)		No: 5 (35.7) Yes: 9 (64.3)		No: 11 (78.6) Yes: 3 (21.4)		Mild: 1 (7.1) Moderate: 2 (14.3) Severe: 10 (71.4) None: 1 (7.1)	
Slow (n = 28)	Daily: 10 (35.7) Free: 6 (21.4) Monthly: 1 (3.6) Sporadic: 10 (35.7) Weekly: 1 (3.6)		No: 9 (32.1) Yes: 19 (67.9)		No: 17 (60.7) Yes: 11 (39.3)		No: 18 (64.3) Yes: 10 (35.7)		Mild: 3 (10.7) Moderate: 3 (10.7) Severe: 13 (46.4) None: 9 (32.1)	
Epileptiform abnormalities										
Focal and multifocal (n = 43)	Sporadic: 14 (32.6) Monthly: 2 (4.7) Weekly: 3 (7.0) Daily: 13 (30.2) Free: 11 (25.6)	0.346	No: 12 (27.9) Yes: 31 (72.1)	0.004*	No: 22 (51.2) Yes: 21 (48.8)	0.008*	No: 29 (67.4) Yes: 14 (32.6)	0.684	Mild: 2 (4.7) Moderate: 9 (20.9) Severe: 24 (55.8) None: 8 (18.6)	0.000*
Generalized (n = 12)	Sporadic: 5 (41.7) Monthly: 1 (8.3) Daily: 5 (41.7) Free: 1 (8.3)		No: 5 (41.7) Yes: 7 (58.3)		No: 10 (83.3) Yes: 2 (16.7)		No: 7 (58.3) Yes: 5 (41.7)		Mild: 5 (41.7) Moderate: 2 (16.7) Severe: 2 (16.7) None: 3 (25.0)	
None (n = 52)	Sporadic: 18 (34.6) Monthly: 7 (13.5) Weekly: 4 (7.7) Daily: 7 (13.5) Free: 16 (30.8)		No: 32 (61.5) Yes: 20 (38.5)		No: 41 (78.8) Yes: 11 (21.2)		No: 37 (71.2) Yes: 15 (28.8)		Mild: 11 (21.2) Moderate: 4 (7.7) Severe: 12 (23.1) None: 25 (48.1)	

Abbreviations: ASD = autistic spectrum disorder; DD = developmental delay; DR = drug resistance; ID = intellectual disability; MD = movement disorder.
*Indicates statistical significance at $p < 0.05$ (95% CI).

respectively) and focal/multifocal epileptiform abnormalities ($r = 0.33$, $r = 0.16$).

Finally, one more positive correlation was identified between *MECP2* genetic variants and both generalized seizures ($r = 0.29$) and epilepsy onset after the first year of life ($r = 0.45$).

The statistical correlation between each combination of clinical and EEG features at onset vs causative genes is presented in Figure 4 ($p < 0.05$ and $p < 0.01$ with 1 and 2 starred squares, respectively).

Figure 5 illustrates a comprehensive schematic representation of the results between clustering of causative genes and both clinical and EEG features.

Discussion

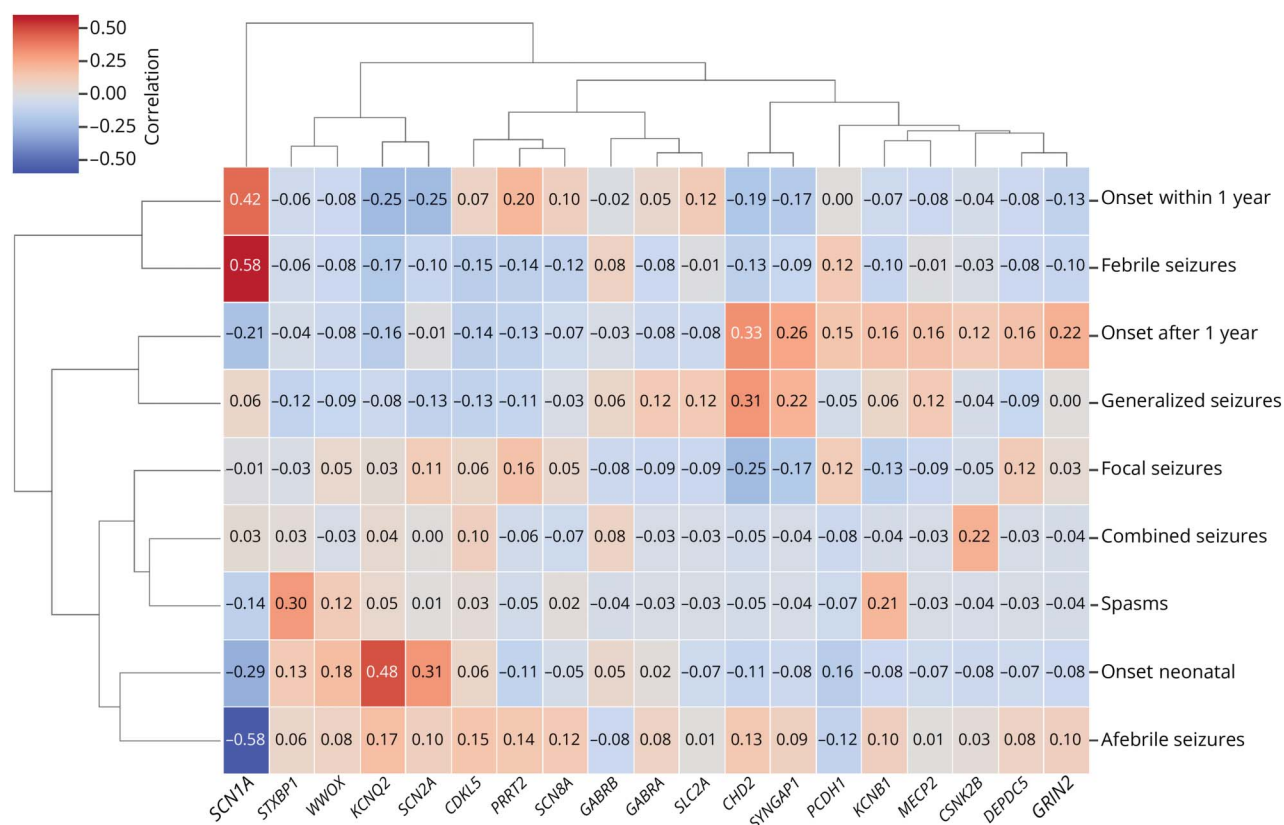
This study analyzed a large cohort of patients with genetic epilepsies using HCA to identify homogeneous subgroups defined by specific genetic causes, each showing distinct clinical and EEG patterns. This method improves understanding of

how epilepsy genes relate to clinical and EEG features, supporting molecular diagnostics and personalized treatment. The study also examined how early clinical and EEG findings at seizure onset are associated with outcomes, helping clinicians better inform prognosis and guide early management decisions.

Owing to the wide variability in phenotypic expression in genetic epilepsies, scattered clinical findings at onset are not sufficiently reliable to hypothesize a definite outcome. To this purpose, we compared clinical and EEG data at epilepsy onset with clinical and genetic data at follow-up using both inferential and hierarchical clustering analyses in 277 patients.

The main results of our study revealed that age at onset, type of seizures, febrile/afebrile seizures, and EEG data were statistically associated with specific clinical features at follow-up. Epilepsy onset after the first year of life was correlated with sporadic seizure frequency at follow-up (41.7%). An earlier onset (<1 year) was most frequently seen in DEEs, where persistence of seizures at follow-up¹⁴ was also more frequent. In our cohort, the age at onset in the neonatal period was correlated with higher incidence of drug resistance (60.7%),

Figure 2 Hierarchical Clustering Analysis Between Clinical Data and Causative Genes



Correlation matrix based on Kendall τ coefficient and hierarchical clustering between clinical data and causative genes. This figure displays the correlation matrix between clinical features (rows) and causative genes (columns), calculated using the Kendall τ coefficient. The numerical values within each cell represent the Kendall τ correlation coefficients, which range from -1 to 1 and reflect the strength and direction of monotonic relationships between variables. The cell colors indicate the direction and magnitude of the correlation: shades of red represent positive correlations, blue indicates negative correlations, and white reflects no correlation. The color scale is shown in the legend (top left), ranging from -0.50 (intense blue) to $+0.50$ (intense red). Both clinical and genes are grouped using hierarchical clustering, with dendrograms shown along the rows and columns to highlight patterns of association. The figure ($n = 226$) shows that *SYNGAP1/CHD2* is associated with generalized seizures and onset after 1 year; *SCN1A* with febrile seizures and onset within 1 year; and *SCN2A/KCNQ2* with neonatal onset.

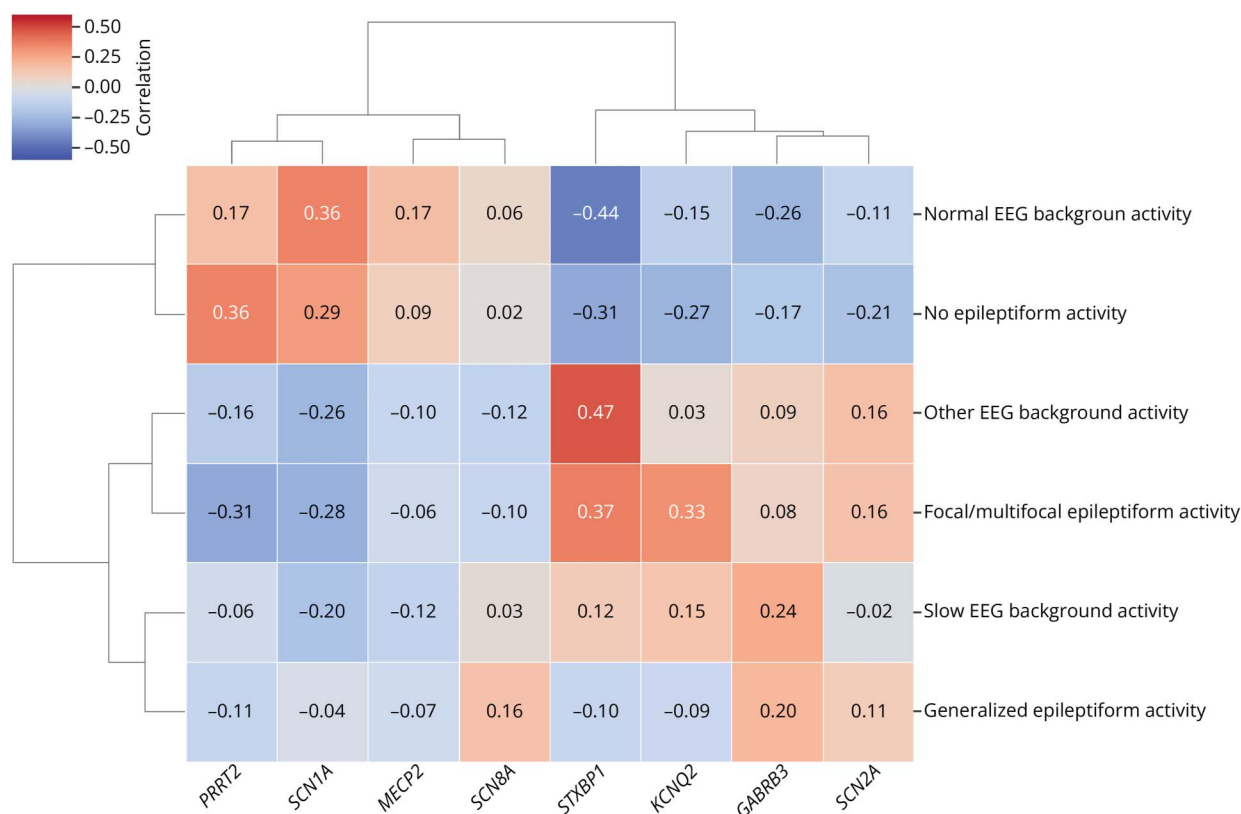
and this has been also pointed out elsewhere.¹⁵ In our sample, neonatal onset was also associated with higher probability of developing movement disorders at follow-up (60.7%). To date, no studies have demonstrated this peculiar association, even if the presence of movement disorders was known to be typical of some genetic early infantile DEEs, such as *STXBP1* or *KCNQ2*-related DEEs.^{16,17} We found that a severe DD/ID at follow-up was more frequently seen in patients with neonatal onset (71.4%), in whom there is a higher probability of developing DEE.¹⁴

The occurrence of afebrile seizures was associated with a worse outcome when compared with febrile seizures. Sixty-three percent of patients with afebrile seizures at onset exhibited drug resistance, and 43.3% had a severe DD/ID at follow-up. It has been demonstrated that, in the presence of a *SCN1A* variant, an earlier onset of afebrile seizures was associated with the development of a Dravet phenotype.¹⁸

EEG data at onset may suggest the prognosis at follow-up. The presence of a slow BA and/or focal and multifocal epileptiform

abnormalities at onset is associated with a higher incidence of drug resistance and with severe DD/ID at follow-up. Specific studies already reported similar findings in patients with DEEs,¹⁹ as well as in those with Aicardi syndrome.²⁰ We highlighted that EEG can be considered as a biomarker of drug resistance and poor cognitive outcome in an unselected population of patients with genetic epilepsies. It is important to note that, even if several of these correlations (such as early-onset epilepsy with drug resistance or slow EEG background with poor cognitive outcome) have been previously reported, our study demonstrated that these correlations retain their validity in a genetically confirmed context, thereby supporting their use in routine prognostication, even when the causal gene has already been detected. Moreover, the strong link we observed between neonatal-onset seizures and subsequent movement disorders highlights a clinical trajectory that is often overlooked. This suggests that these patients may benefit from earlier counseling and more tailored management strategies.

We used HCA to identify homogeneous subgroups of genetic etiologies in which clinical and EEG characteristics were thus

Figure 3 Hierarchical Clustering Analysis Between EEG Data and Causative Genes

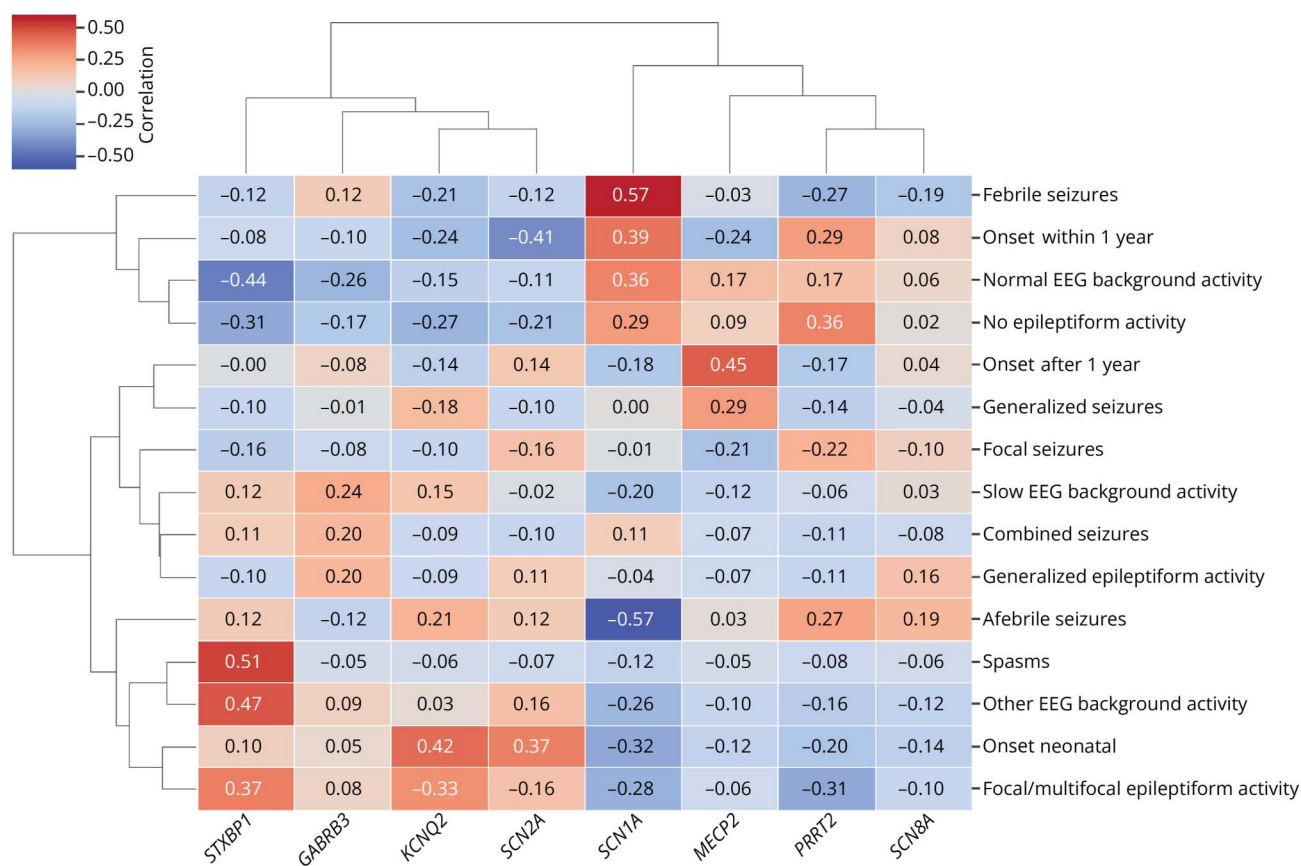
Correlation matrix based on Kendall τ coefficient and hierarchical clustering between EEG data and causative genes. This figure displays the correlation matrix between EEG data (rows) and causative genes (columns), calculated using the Kendall τ coefficient. The numerical values within each cell represent the Kendall τ correlation coefficients, which range from -1 to 1 and reflect the strength and direction of monotonic relationships between variables. The cell colors indicate the direction and magnitude of the correlation: shades of red represent positive correlations, blue indicates negative correlations, and white reflects no correlation. The color scale is shown in the legend (top left), ranging from -0.50 (intense blue) to +0.50 (intense red). Both EEG data and genes are grouped using hierarchical clustering, with dendrograms shown along the rows and columns to highlight patterns of association. The figure ($n = 61$) shows that *SCN1A* and *PRRT2* are linked to normal background EEG activity and absence of epileptiform abnormalities and *STXBP1* is correlated with other background EEG activity and focal/multifocal epileptiform activity.

identified. Clustering methods supported clinicians in identifying and correlating different seizure semiology subgroups with anatomic organization in patients with frontal lobe epilepsy.²¹ HCA was also used to identify different phenotypic groups associated with infratentorial morphologic MRI patterns in spinocerebellar ataxia spectrum.²² Clinical, neuroimaging, and genetic data analyzed using HCA allowed identification of different phenotypic subgroups of patients with a diagnosis of mitochondrial encephalomyopathy, lactic acidosis, and stroke-like episodes.²³ Moreover, in patients with tuberous sclerosis complex, hierarchical cluster analysis applied to neuropsychiatric evaluations has identified distinct subgroups of tuberous sclerosis complex-associated neuropsychiatric disorders.²⁴

Our analysis documented several associations between specific groups of clinical features at onset and causative genes. In patients with age at onset after first year and with occurrence of generalized seizures, the most likely etiology was related to *CHD2*/*SYNGAP1* genetic variants. The frequency of such a clinical presentation for *CHD2* and *SYNGAP1* has already been reported.^{25,26} In particular, de novo variants in *CHD2* are

associated with variable epilepsy and neurodevelopmental disorders.²⁵ Pathogenic *CHD2* variants have been described in patients with epilepsy with myoclonic-atonic seizure (EMaTS), although several patients exhibited a developmental delay before seizure onset, not fully consistent with a classic EMaTS presentation.²⁷ Pathogenic de novo variants in *SYNGAP1* may cause an epileptic phenotype of “atypical” EMaTS also characterized by the presence of hypotonia, DD/ID, and movement disorders.²⁸ The co-clustering of *CHD2* and *SYNGAP1* reflects convergence at the phenotypic level rather than shared molecular function, as *CHD2* acts in chromatin remodeling during cortical development, whereas *SYNGAP1* regulates excitatory synaptic signaling. Both genes, however, are associated with early developmental delay, generalized epilepsy, and photosensitivity.^{29,30}

Febrile seizures and onset within the first year of life clustered with *SCN1A* genetic variants, as reported.³¹ The clustering analysis including clinical and EEG data at onset vs causative gene revealed that *SCN1A* genetic variants clustered with onset within first year, febrile seizures, normal BA, and absence of epileptiform abnormalities. Pathogenic variants in

Figure 4 Hierarchical Clustering Analysis Between Combined Clinical/EEG Data and Causative Genes

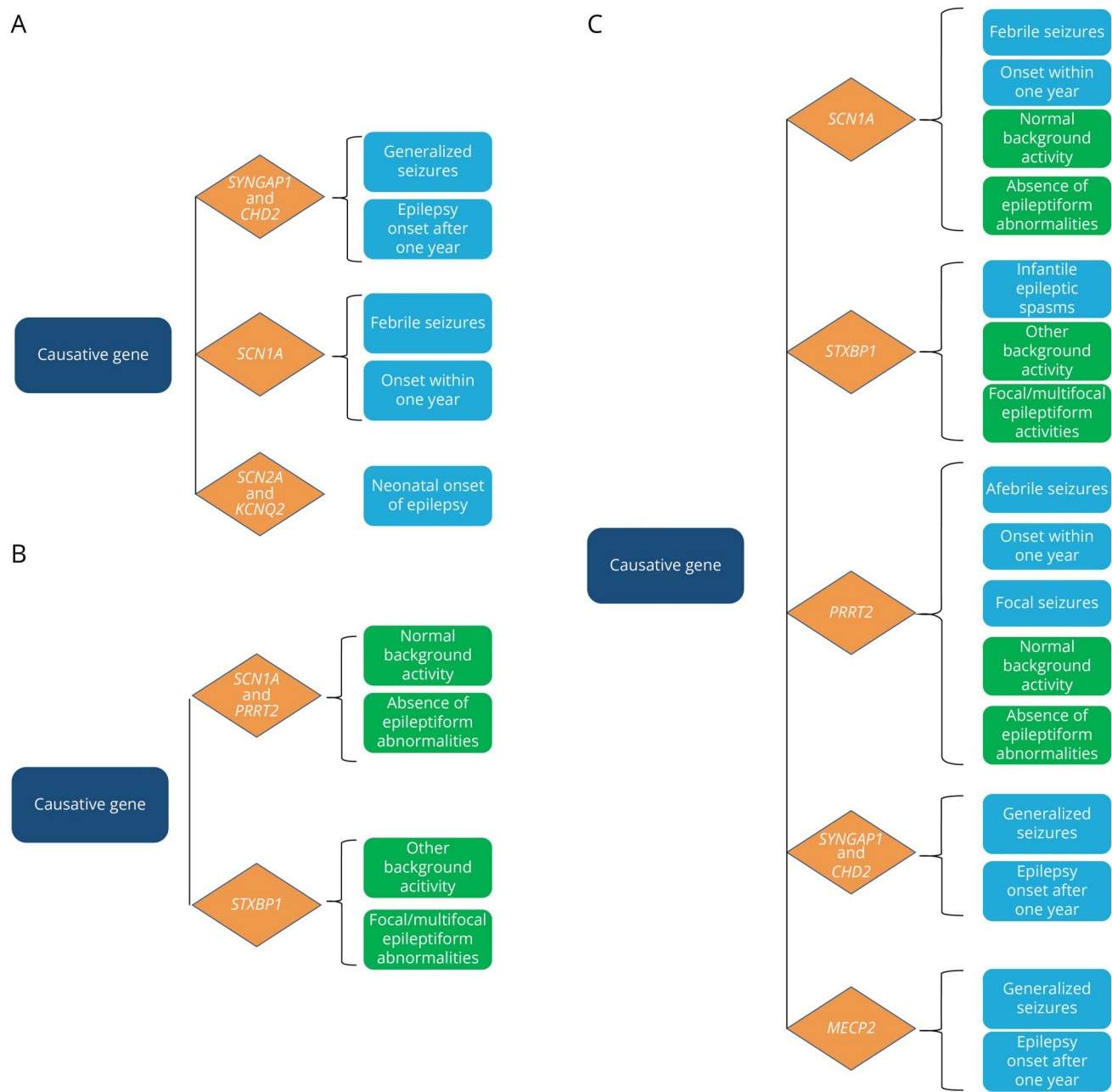
Correlation matrix based on Kendall τ coefficient and hierarchical clustering between combined clinical/EEG data and causative genes in individuals with epilepsy. This figure displays the correlation matrix between a specific combination of clinical and EEG features (rows) and a panel of causative genes (columns), calculated using the Kendall τ coefficient. Each cell shows the Kendall τ value, ranging from -1 to 1, which reflects the strength and direction of the monotonic association between gene variants and specific clinical or neurophysiologic characteristics. The cell colors indicate the direction and magnitude of the correlation: shades of red represent positive correlations, blue indicates negative correlations, and white reflects no correlation. The color scale is shown in the legend (top left), ranging from -0.50 (intense blue) to +0.50 (intense red). Both clinical/EEG data and genes were grouped using hierarchical clustering, allowing the identification of association patterns and phenotype clusters. Notable associations include the following: *SCN1A* was associated with febrile seizures and seizure onset within the first year of life, *SCN2A* and *KCNQ2* were associated with neonatal onset, *SYNGAP1* and *CHD2* were linked to generalized seizures and seizure onset after 1 year, *STXBP1* had positive correlations with abnormal EEG background activity and focal/multifocal epileptiform discharges, and *SCN1A* and *PRRT2* were associated with normal EEG background and absence of epileptiform abnormalities.

SCN1A are associated with a broad spectrum of epilepsy syndromes.³¹ Among these, the most frequent are forms in which epilepsy is predominantly triggered by fever, especially at onset, and include mild phenotypes encompassing the GEFS+ spectrum and severe phenotypes as in DS. DS and GEFS+ are caused by LOF *SCN1A* variants. Studies on cellular models of both DS and GEFS+ showed a marked reduction in the action potential firing in GABAergic pathologic neurons and an increased activity of glutamatergic neurons causing hyperexcitability.³² Recently, other less common epilepsy phenotypes were reported, including neonatal-onset DEE and focal epilepsy, due to GOF *SCN1A* variants.³³ Epilepsy onset in the first year of life, characterized by fever triggered seizures, is also seen in other genetic epilepsies. The main differential diagnosis is *PCDH19*-clustering epilepsy, an X-linked condition mainly affecting female individuals. Beyond the predominance of female individuals, *PCDH19*-related epilepsy is characterized by a later age at onset, seizures that are typically brief and clustered, and infrequent clonic and

hemiclonic seizures. Instead, no significant differences were found in EEG features, with poor interictal abnormalities and a primarily normal BA in both conditions at onset.³⁴ The early recognition of *SCN1A*-related epilepsies, particularly DS, has significant therapeutic implications. The use of fenfluramine before the age of 2 years emerged as a promising opportunity for seizure control and for potential protection of the neurodevelopmental outcome.³⁵ In addition, preclinical studies on murine models have shown promising results regarding the use of antisense oligonucleotides (ASOs) in the treatment of the disease, demonstrating restoration of the action potential firing, sodium current density, and GABAergic signaling in DS interneurons,³⁶ and promising results have also been reported in phase 2 studies.

PRRT2 gene variants clustered with normal BA, absence of epileptiform abnormalities, and age at onset within the first year and were independently correlated with the presence of focal and afebrile seizures. *PRRT2* variants are associated with

Figure 5 Associations Between Clustering Results of Causative Genes and Both Clinical and EEG Data



Comprehensive schematic representation of the associations between clustering results of causative genes and both clinical and EEG data. The diamond-shaped orange nodes represent individual causative genes or gene clusters. These genes or clusters are connected to either blue or green rectangular boxes, corresponding respectively to clinical features and EEG data. (A) Clinical data and causative gene associations. (B) EEG data and causative gene correlations. (C) Integration of clinical and EEG data and causative gene.

a spectrum of neurologic overlapping conditions, including epilepsy, movement disorder, and migraine. The most frequent epilepsy phenotype is self-limited infantile epilepsy (SeLIE), with epilepsy onset in the first year.³⁷ In SeLIE, seizures are almost always afebrile, although seizures triggered by fever (in the spectrum of febrile seizures or GEFS+) are rarely reported.^{38,39} Less frequently, epilepsy onset can occur in childhood, mainly presenting a childhood absence epilepsy phenotype.⁴⁰ In epilepsies with an onset in the first year, focal seizures, and normal BA, a normal interictal EEG could be

indicative of a genetic etiology, as in SeLIE,^{41,42} whereas interictal localized focal spikes suggests structural epilepsy.⁴³

We also performed HCA looking at EEG data alone vs causative genes, and we found an interesting correlation between *SCN1A* and *PRRT2* variants with normal BA and absence of interictal epileptiform abnormalities. Thus, when considering only EEG data at seizure onset, we observed a similarity between these 2 etiologies. However, this correspondence is lost when clinical data are incorporated.

Although the age at onset is approximately overlapping in both conditions, only the presence of fever (predominantly associated with *SCN1A* variants) prevents clustering between the 2 conditions.

STXBP1 variants clustered with epileptic spasms and the presence of burst suppression/hypsarrhythmia/paroxysmal alternating background EEG activity and independently with focal and multifocal epileptiform abnormalities. *STXBP1*-related DEE is among the most common genetic epilepsies.⁴⁴ Approximately half of the affected individuals have epileptic spasms, especially in the presence of truncating variants and deletions when compared with missense variants.^{44,45} Most patients have focal or multifocal epileptiform discharges, regardless of seizure semiology, and burst suppression and hypsarrhythmia are frequent EEG findings.⁴⁶

KCNQ2 and *SCN2A* clustered with neonatal-onset epilepsy and focal/multifocal EEG epileptiform abnormalities. It should be noted that our cohort includes both DEEs and SeLNE, with a prevalence of DEEs. This discrepancy could explain the association between neonatal onset and the presence of focal/multifocal interictal abnormalities, which are more frequent in DEEs than in self-limited epilepsies.^{47,48} The age at onset and the clinical presentation are influenced by the functional effect of the genetic variant. An early onset in *SCN2A*-related epilepsy is associated with the presence of GOF variants, whereas a later onset is related to LOF variants or missense variants with mixed GOF and LOF effect.⁴⁹ At the same time, although LOF variants in *KCNQ2* are associated with neonatal-onset epilepsy, GOF variants cause a more complex phenotype, characterized by neonatal encephalopathy, nonepileptic myoclonus, DD, and infantile onset of seizures, often with epileptic spasms.⁵⁰ When only EEG data were evaluated in the HCA, we did not observe a clustering between *SCN2A* and *KCNQ2*, which instead occurs if clinical data are also included. This finding may suggest a lower specificity of EEG data alone in the correlation between these 2 genetic etiologies. *SCN2A* and *KCNQ2* cluster represents a coherent group, with both genes modulating neuronal excitability through voltage-gated ion channel function, consistent with their shared association with neonatal-onset epileptic encephalopathy.

MECP2 clustered with onset after first year and the occurrence of generalized seizures. Late-onset seizures in Rett syndrome were already known, whereas no specific pattern or type of seizures was reported.⁵¹

This study has limitations, including the small number of patients per gene and limited EEG data at epilepsy onset, due to its retrospective design. The lack of some EEG data reduced the heterogeneity of the sample. Therefore, future studies should extend the analyzed cohort to confirm the encouraging results obtained from this first HCA approach. This could also allow analysis of the correlation between clinical and EEG data and specific genetic variants for each causative gene, based on the type of variant, functional domains involved, and predicted

functional effect. It would also have been important to assess psychomotor and intellectual development at onset in our cohort; this was not possible because of the lack of homogeneous and structured assessments at disease onset in many patients and should be considered in future studies. While current recommendations already support the use of large gene panels or exome sequencing for the diagnosis of genetic epilepsies, our HCA provides complementary clinical value. By showing how different genes converge into specific phenotype clusters, HCA may support early clinical reasoning before genetic confirmation is available, facilitating interim management choices and counseling of families.

Our study represents an early application of HCA to investigate the correlation between clinical, EEG data and causative genes in a large cohort of patients with genetic epilepsy.

Our study highlights the complementary value of integrating clinical and EEG features at seizure onset. Although analyses based on either clinical or EEG data alone largely reproduced known associations, the combined analysis enhanced specificity by generating more distinct and clinically recognizable clusters. This may suggest that an integrated approach may offer greater discriminatory power for early diagnostic reasoning and clinical application.

The application of HCA to combined clinical and EEG features provides a novel framework that may enhance early diagnostic reasoning, guide interim treatment decisions, and inform the design of targeted clinical trials. Our results add a new value to the existing literature and propose new method for translating early phenotypic markers into clinical practice.

Early clinical recognition of genetic epilepsies is essential, as it allows the use of targeted therapies and prevents ineffective or harmful treatments. Timely diagnosis also enables more effective drug combinations and prepares the way for emerging treatments such as ASOs and gene therapies. Even in settings with limited genetic testing, early identification remains crucial to improve outcomes and support personalized care.

In our opinion, HCA may play a crucial role in the study of rare diseases, adding a new support to the deep phenotyping and the genotype-phenotype correlations, even disclosing mild clinical variabilities within the same epilepsy syndrome. This approach could help identify specific disease trajectories, support the interpretation of long-term outcomes, and provide insight into the natural course of the disease. It may also help identify diagnostic or prognostic biomarkers, assess treatment responses across patient subgroups, and guide the design of clinical trials by enabling the selection of more homogeneous patient cohorts—ultimately improving the likelihood of detecting meaningful treatment effects.

Author Contributions

A. De Dominicis: drafting/revision of the manuscript for content, including medical writing for content; major role in

the acquisition of data; study concept or design; analysis or interpretation of data. M. Mercier: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; analysis or interpretation of data. G. Carfi Pavia: major role in the acquisition of data. L.M. Piscitello: major role in the acquisition of data. S. Capelletti: major role in the acquisition of data. C. Filosomi: major role in the acquisition of data. C. Correale: major role in the acquisition of data. C. Calabrese: major role in the acquisition of data. L. De Palma: drafting/revision of the manuscript for content, including medical writing for content; study concept or design. A. Novelli: drafting/revision of the manuscript for content, including medical writing for content. F. Stregapede: drafting/revision of the manuscript for content, including medical writing for content. M. Trivisano: drafting/revision of the manuscript for content, including medical writing for content; study concept or design. N. Specchio: drafting/revision of the manuscript for content, including medical writing for content.

Study Funding

This study was supported by #NEXTGENERATIONEU (NGEU) and funded by the Ministry of University and Research (MUR), National Recovery and Resilience Plan (NRRP), project MNESYS (PE0000006)—a multiscale integrated approach to the study of the nervous system in health and disease (DN. 1553 11.10.2022). This study was also funded by Ministry of Health with Current research Funds.

Disclosure

M. Trivisano has received speaker fees or fundings or has participated in advisory boards for BioMarin and Biocodex. N. Specchio has served on scientific advisory boards for GW Pharma, BioMarin, Arvelle, Marinus and Takeda; has received speaker honoraria from Eisai, Biomarin, Livanova, Sanofi; and has served as an investigator for Zogenix, Marinus, Biomarin, UCB, Roche. The remaining authors have no conflicts of interest. Go to Neurology.org/N for full disclosures.

Publication History

Received by *Neurology*® August 28, 2025. Accepted in final form December 23, 2025. Submitted and externally peer reviewed. The handling editor was Associate Editor Emily Johnson, MD, MPH.

References

- Symonds JD, Zuberi SM, Stewart K, et al. Incidence and phenotypes of childhood-onset genetic epilepsies: a prospective population-based national cohort. *Brain*. 2019; 142(8):2303-2318. doi:10.1093/brain/awz195
- Oliver KL, Scheffer IE, Bennett MF, Grinton BE, Bahlo M, Berkovic SF. Genes4Epilepsy: an epilepsy gene resource. *Epilepsia*. 2023;64(5):1368-1375. doi:10.1111/epi.17547
- Scheffer IE, Zuberi S, Mefford HC, Guerrini R, McTague A. Developmental and epileptic encephalopathies. *Nat Rev Dis Prim*. 2024;10(1):61. doi:10.1038/s41572-024-00546-6
- Specchio N, Trivisano M, Aronica E, et al. The expanding field of genetic developmental and epileptic encephalopathies: current understanding and future perspectives. *Lancet Child Adolesc Health*. 2024;8(11):821-834. doi:10.1016/S2352-4642(24)00196-2
- Baer S, Schalk A, Miguët M, et al. Copy number variation and epilepsy: state of the art in the era of high-throughput sequencing: a multicenter cohort study. *Pediatr Neurol*. 2024;159:16-25. doi:10.1016/j.pediatrneurol.2024.07.007
- Guerrini R, Conti V, Mantegazza M, Balestrini S, Galanopoulou AS, Benfenati F. Developmental and epileptic encephalopathies: from genetic heterogeneity to phenotypic continuum. *Physiol Rev*. 2023;103(1):433-513. doi:10.1152/physrev.00063.2021
- Millervet C, Weckhuysen S; ILAE Genetics Commission. ILAE genetic literacy series: self-limited familial epilepsy syndromes with onset in neonatal age and infancy. *Epileptic Disord*. 2023;25(4):445-453. doi:10.1002/epd2.20026
- Biervert C, Steinlein OK. Structural and mutational analysis of KCNQ2, the major gene locus for benign familial neonatal convulsions. *Hum Genet*. 1999;104(3):234-240. doi:10.1007/pl00008713
- Goto A, Ishii A, Shibata M, Ihara Y, Cooper EC, Hirose S. Characteristics of KCNQ2 variants causing either benign neonatal epilepsy or developmental and epileptic encephalopathy. *Epilepsia*. 2019;60(9):1870-1880. doi:10.1111/epi.16314
- Brunklaus A, Pérez-Palma E, Ghanty I, et al. Development and validation of a prediction model for early diagnosis of SCN1A-related epilepsies. *Neurology*. 2022; 98(11):E1163-E1174. doi:10.1212/WNL.0000000000200028
- Li Y, Su Z, Zhao K, et al. Epigenetic mechanisms in the pathophysiology and progression of epilepsy: a comprehensive review of experimental and clinical studies. *Curr Neuropsychopharmacol*. 2025;23(9):1047-1059. doi:10.2174/1570159X23666241220163832
- Oliver KL, Ellis CA, Scheffer IE, et al. Common risk variants for epilepsy are enriched in families previously targeted for rare monogenic variant discovery. *EBioMedicine*. 2022;81:104079. doi:10.1016/j.ebiom.2022.104079
- Little J, Higgins JP, Ioannidis JP, et al. *STrengthening the REporting of Genetic Association Studies (STREGA): An Extension of the STROBE Statement*. *Genet Epidemiol*. 2009;33(7):S81-98. doi: 10.1002/gepi.20410
- Poke G, Stanley J, Scheffer IE, Sadleir LG. Epidemiology of developmental and epileptic encephalopathy and of intellectual disability and epilepsy in children. *Neurology*. 2023;100(13):E1363-E1375. doi:10.1212/WNL.0000000000206758
- Perucca E, Perucca P, White HS, Wirrell EC. Drug resistance in epilepsy. *Lancet Neurol*. 2023;22(8):723-734. doi:10.1016/S1474-4422(23)00151-5
- Milh M, Villeneuve N, Chouchane M, et al. Epileptic and nonepileptic features in patients with early onset epileptic encephalopathy and STXBP1 mutations. *Epilepsia*. 2011;52(10):1828-1834. doi:10.1111/j.1528-1167.2011.03181.x
- Pisano T, Numis AL, Heavin SB, et al. Early and effective treatment of KCNQ2 encephalopathy. *Epilepsia*. 2015;56(5):685-691. doi:10.1111/epi.12984
- de Lange IM, Gunning B, Sonnsma ACM, et al. Influence of contraindicated medication use on cognitive outcome in Dravet syndrome and age at first afebrile seizure as a clinical predictor in SCN1A-related seizure phenotypes. *Epilepsia*. 2018;59(6):1154-1165. doi:10.1111/epi.14191
- Surdi P, Trivisano M, De Dominicis A, et al. Unveiling the disease progression in developmental and epileptic encephalopathies: insights from EEG and neuropsychology. *Epilepsia*. 2024;65(11):3279-3292. doi:10.1111/epi.18127
- Masnada S, Alfei E, Formica M, et al. EEG at onset and MRI predict long-term clinical outcome in Aicardi syndrome. *Clin Neurophysiol*. 2022;142:112-124. doi:10.1016/j.clinph.2022.07.496
- Bonini F, McGonigal A, Trébouchon A, et al. Frontal lobe seizures: from clinical semiology to localization. *Epilepsia*. 2014;55(2):264-277. doi:10.1111/epi.12490
- Zhao J, Zeng W, Huang X, et al. Analysis and hierarchical clustering of infratentorial morphological MRI identifies SCAs phenogroups. *J Neurol*. 2023;270(9):4466-4477. doi:10.1007/s00415-023-11792-1
- Alves CAPE, Zandifar A, Peterson JT, et al. MELAS: phenotype classification into classic-versus-atypical presentations. *AJNR Am J Neuroradiol*. 2023;44(5):602-610. doi:10.3174/ajnr.A7837
- De Vries PJ, Belousova E, Benedik MP, et al; TOSCA Consortium and TOSCA Investigators. Natural clusters of tuberous sclerosis complex (TSC)-associated neuropsychiatric disorders (TAND): new findings from the TOSCA TAND research project. *J Neurodev Disord*. 2020;12:24. doi:10.1186/s11689-020-09327-0
- Carvill GL, Mefford HC. CHD2-related neurodevelopmental disorders. In: Adam MP, Bick S, Mirzaa GM, Pagon RA, Wallace SE, Amemiya A, eds. *GeneReviews*® [Internet]. University of Washington, Seattle; 2015:1993–2026. Updated January 21, 2021.
- Holder JL Jr, Hamdan FF, Michaud JL. SYNGAP1-related intellectual disability. In: Adam MP, Bick S, Mirzaa GM, Pagon RA, Wallace SE, Amemiya A, eds. *GeneReviews*® [Internet]. University of Washington, Seattle; 2019: 1993–2026. Updated August 14, 2025.
- Trivisano M, Striano P, Sartorelli J, et al. CHD2 mutations are a rare cause of generalized epilepsy with myoclonic-astatic seizures. *Epilepsy Behav*. 2015;51:53-56. doi:10.1016/j.yebeh.2015.06.029
- Tang S, Addis L, Smith A, et al. Phenotypic and genetic spectrum of epilepsy with myoclonic atonic seizures. *Epilepsia*. 2020;61(5):995-1007. doi:10.1111/epi.16508
- Puri AG, Woods S, Schreiber JM, et al. Characterizing CHD2-associated epilepsy: a multicenter study and pooled analysis of the literature. *Seizure*. 2025;132:88-97. doi:10.1016/j.seizure.2025.09.004
- Greco MR, Chatterjee M, Taylor AM, Gropman AL. SYNGAP1 syndrome and the Brain Gene Registry. *Genes (Basel)*. 2025;16(4):405. doi:10.3390/genes16040405
- Scheffer IE, Nabbout R. SCN1A-related phenotypes: Epilepsy and beyond. *Epilepsia*. 2019;60(suppl 3):S17-S24. doi:10.1111/epi.16386
- Zayat V, Szlendak R, Hoffman-Zacharska D. Concise review: stem cell models of SCN1A-related encephalopathies: current perspective and future therapies. *Cells*. 2022;11(19):3119. doi:10.3390/cells11193119
- Matricardi S, Cestè S, Trivisano M, et al. Gain of function SCN1A disease-causing variants: expanding the phenotypic spectrum and functional studies guiding the choice of effective antiseizure medication. *Epilepsia*. 2023;64(5):1331-1347. doi:10.1111/epi.17509
- Trivisano M, Pietrafusa N, di Ciommo V, et al. PCDH19-related epilepsy and Dravet syndrome: face-off between two early-onset epilepsies with fever sensitivity. *Epilepsy Res*. 2016;125:32-36. doi:10.1016/j.eplepsyres.2016.05.015

35. Pietrafusa N, Trivisano M, Casellato S, et al. Fenfluramine below the age of 2 years in Dravet syndrome: what about safety and efficacy? *Epilepsia*. 2024;65:e7-e13. doi: 10.1111/epi.17854
36. Yuan Y, Lopez-Santiago L, Denomme N, et al. Antisense oligonucleotides restore excitability, GABA signalling and sodium current density in a Dravet syndrome model. *Brain*. 2024;147(4):1231-1246. doi:10.1093/brain/awad349
37. Lee J, Kim YO, Lim BC, Lee J. PRRT2-positive self-limited infantile epilepsy: initial seizure characteristics and response to sodium channel blockers. *Epilepsia Open*. 2023; 8(2):436-443. doi:10.1002/epi4.12708
38. Scheffer IE, Grinton BE, Heron SE, et al. PRRT2 phenotypic spectrum includes sporadic and fever-related infantile seizures. *Neurology*. 2012;79(21):2104-2108. doi: 10.1212/WNL.0b013e3182752c6c
39. Zheng W, Zhang J, Deng X, et al. Identification of a premature termination mutation in the proline-rich transmembrane protein 2 gene in a Chinese family with febrile seizures. *Mol Neurobiol*. 2016;53:835-841. doi:10.1007/s12035-014-9047-4
40. Yang K, Quiroz V, Ebrahimi-Fakhari D. PRRT2-related disorder. 2018. In: Adam MP, Bick S, Mirzaa GM, Pagon RA, Wallace SE, Amemiya A, eds. *GeneReviews*[®] [Internet]. University of Washington, Seattle; 2018: 1993–2026. Updated July 4, 2024.
41. Miceli F, Soldovieri MV, Weckhuysen S, Cooper E, Tagliatela M. KCNQ2-related disorders. In: Adam MP, Bick S, Mirzaa GM, Pagon RA, Wallace SE, Amemiya A, eds. *GeneReviews*[®] [Internet]. University of Washington, Seattle; 2010:1993–2026. Updated May 19, 2022.
42. Hammer MF, Xia M, Schreiber JM. SCN8A-related epilepsy and/or neurodevelopmental disorders. In: Adam MP, Bick S, Mirzaa GM, Pagon RA, Wallace SE, Amemiya A, eds. *GeneReviews*[®] [Internet]. University of Washington, Seattle; 2016: 1993–2026. Updated April 6, 2023.
43. Ramakrishnan S, Asuncion RMD, Rayi A. Localization-related epilepsies on EEG. In: *StatPearls* [Internet]. StatPearls Publishing; 2025.
44. Xian J, Parthasarathy S, Ruggiero SM, et al. Assessing the landscape of STXBP1-related disorders in 534 individuals. *Brain*. 2022;145(5):1668-1683. doi:10.1093/brain/awab327
45. Thalwitzer KM, Xian J, de Campo D, et al. Early life seizures and epileptic spasms in STXBP1-related disorders. *Epilepsia*. 2024;65(3):805-816. doi:10.1111/epi.17886
46. Stamberger H, Nikanorova M, Willemsen MH, et al. STXBP1 encephalopathy. *Neurology*. 2016;86(10):954-962. doi:10.1212/WNL.0000000000002457
47. Xu Y, Dou Y, Chen X, et al. Early initial video-electro-encephalography combined with variant location predict prognosis of KCNQ2-related disorder. *BMC Pediatr*. 2021;21:477. doi:10.1186/s12887-021-02946-z
48. Wolff M, Johannesen KM, Hedrich UBS, et al. Genetic and phenotypic heterogeneity suggest therapeutic implications in SCN2A-related disorders. *Brain*. 2017;140(5): 1316-1336. doi:10.1093/brain/awx054
49. Reynolds C, King MD, Gorman KM. The phenotypic spectrum of SCN2A-related epilepsy. *Eur J Paediatr Neurol*. 2020;24:117-122. doi:10.1016/j.ejpn.2019.12.016
50. Bayat A, Iavarone S, Miceli F, et al. Phenotypic and functional assessment of two novel KCNQ2 gain-of-function variants Y141N and G239S and effects of amitriptyline treatment. *Neurotherapeutics*. 2024;21(1):e00296. doi:10.1016/j.neuro.2023.10.006
51. Operto FF, Mazza R, Pastorino GMG, Verrotti A, Coppola G. Epilepsy and genetic in Rett syndrome: a review. *Brain Behav*. 2019;9(5):e01250. doi:10.1002/brb3.1250